

NORMAN BUI

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EDUCATION

Carleton University | *Bachelor of Computer Science (Honours), 3.9/4.0 GPA*

Graduated Spring 2026

- Coursework: Quality Assurance, Operating Systems, Data Structures & Algorithms, Database Management Systems, Web Development

EXPERIENCE

Electronic Arts (EA) | *Software Developer Intern*

May 2025 - Aug 2025

- Improved CI/CD load-test reliability from **35% to 100%** by fixing auth failures and unstable workflows
- Developed load, soak, and regression plans for live game services across **10+** feature integrations
- Triaged load-test failures by isolating auth, environment, and feature regressions in CI
- Documented reproducible failure cases and validation notes for automated playtesting workflows

Autodesk | *Software Developer Intern*

Jan 2025 - Apr 2025

- Built TypeScript/Java validation and repair infrastructure for Fusion manufacturing data, including an **O(n)** repair algorithm that improved throughput **8x**
- Deployed GraphQL validation and repair endpoints with **100%** Mocha, Chai, and Sinon coverage
- Fixed permission and error-handling gaps in sensitive manufacturing data workflows
- Expanded endpoint test cases for malformed inputs, authorization failures, and repair edge cases

Kongsberg Geospatial | *Software Developer Intern*

May 2024 - Aug 2024

- Built client-facing C++/Qt geospatial demos for aviation, defense, and government use cases
- Integrated LiDAR, WMTS, and S-57 GIS data into real-time mapping and tracking workflows
- Validated geospatial ingestion/rendering edge cases across LiDAR, WMTS, and S-57 demos
- Created repeatable demo checks for map-layer loading, object tracking, and rendering performance

Health Canada | *Data Scientist Intern*

Sep 2023 - Dec 2023

- Architected TDD scraping framework with unittest, expanding sources by **113%** with **100%** coverage
- Validated scraper reliability across schema changes and fault-tolerant parsing paths
- Rebuilt chemical search with BS4/lxml error handling, raising query success by **20%**

Correctional Service Canada | *Software Developer Intern*

May 2022 - Apr 2023

- Improved C# and .NET WET accessibility for WCAG 2.1 Level AA compliance
- Resolved ASP.NET UI defects in victim services app for **100+** daily users
- Supported QA validation for accessible government web workflows
- Verified keyboard navigation, screen-reader labels, and responsive UI behavior after fixes

PROJECTS

Orbital Refueling Simulator | *Python, Streamlit, scikit-learn, pytest*

github.com/nbui23/orbital-refueling

- Modeled **410-second** refueling mission across **9** phases, **9** anomaly classes, and deterministic generation
- Separated hard-limit engineering rules from **9** phase-aware IsolationForest drift models
- Wrote pytest coverage across **5** surfaces: telemetry generation, rules, scoring, phase handling, and explanations

CruxOS | *Next.js, TypeScript, Prisma, SQLite, Expo, React Native*

github.com/nbui23/cruxos

- Built **2-client** CruxOS bouldering app with Expo session capture and Next.js analysis dashboard
- Implemented deterministic insight engine translating **3** signal groups into training and recovery recommendations
- Wrote unit and API integration tests for insight logic, auth flows, and Prisma/SQLite persistence
- Covered typed data contracts across mobile/web capture, review, and SQLite persistence

SKILLS

Languages: Python, TypeScript, JavaScript, C++, C#, Java, SQL, HTML/CSS

Frameworks & Runtimes: Node.js, Express.js, Qt, Spring Boot, Flask, Streamlit, .NET, ASP.NET

AI / ML / Data: LLMs, scikit-learn, pandas, NumPy

Databases & APIs: PostgreSQL, MySQL, SQLite, MongoDB, Firebase, Neo4j, GraphQL

Tools & Practices: Git, AWS, CI/CD, Docker, Docker Compose, Ubuntu, TDD, pytest, Mocha, Chai, Sinon, unittest, BS4, Selenium, Cucumber, PowerShell, Postman, Agile, SDLC

Spoken Languages: English, French, Vietnamese